RoboCup Autonomous
Robot Manipulation (ARM) Challenge

Supported by MathWorks

Participation Guide

Challenge Overview

Problem: In this challenge, your goal is to program an intelligent robot manipulator to identify and sort objects using vision data in dynamic virtual environments.

Prizes:

- **Top teams:** Invited to participate in the RoboCup 2026 Finals using an in-person setup with real Universal Robots UR5e
- **Top 4 international teams:** Qualify for travel support of $1250 to the **RoboCup Finals in Incheon, South Korea**
- **All teams:** Certificates of participation and skills developed in robotics manipulation algorithm development for all team members

Simulation Environment: The manipulation environment used for this challenge is built using Simulink 3D Animation. The environment will be contained in Simulink with the visualization run with Unreal Engine. The robot manipulator used is a Universal UR5e. It is equipped with an RGB camera and a depth camera which can both be used when solving the challenge.

Rules:

- Participants must register in a RoboCup league
- Submissions must contain reproducible MATLAB and/or Simulink files. Any necessary file dependencies must be included.
- Robot perception algorithm must not use environment information available from Simulink 3D Animation
- Any movement induced to the game objects must be a reaction to interactions with other environments components (Do not use Simulink blocks to set positions or states) Modifications to the virtual worlds provided is not suggested as code will be evaluated under the same conditions provided in test environments.

Scoring

- Your model will be scored using the scoring rubric found on the last page. During the semifinals, your submission will be graded on an **unknown 3rd world** not provided in the test environments.
- Top submissions from the qualification round will compete in the finals - During the finals, your submission will be scored based on an in-person setup at the RoboCup finals, that will feature similar objects. Any new conditions will be provided for the qualifying teams.
- Objects will not be reset during judging - If an object is knocked over, your algorithm should account for this.

Awards:

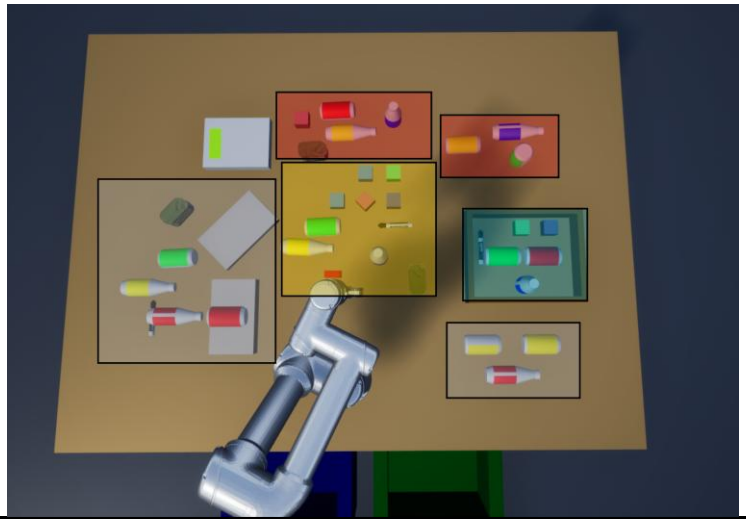
- **Top teams:** Invited to participate in the RoboCup 2026 Finals using an in-person setup with real Universal Robots UR5e
- **Top 4 international teams:** Qualify for up to $1250 travel support to the **RoboCup Finals in Incheon, South Korea**
- **All teams:** Certificates of participation and skills developed in robotics manipulation algorithm development for all team members

Important Dates:

Registration Opens	January 2026
New game environment released	February 19th
Live Challenge Kickoff and Q&A	March 13th
Registration deadline	April 24th
Submission deadline for semifinals round	May 8th (7pm EST)
Announcement of teams qualified for finals	May 11-13
In-person finals (Incheon, South Korea)	June 30 - July 6

Item Layout Conditions

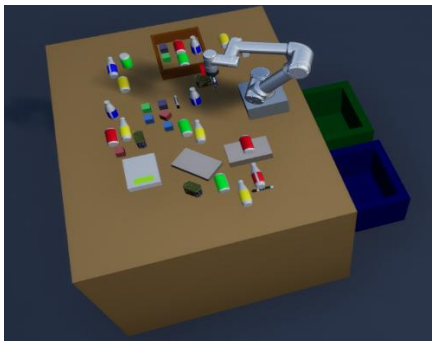
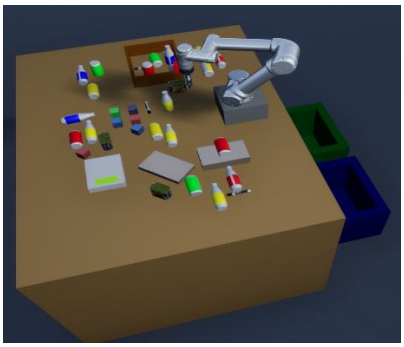
Items present in the virtual world will follow the conditions outlined below in the unknown environment that will be used for scoring.



	Won't change – Items will remain the same. No changes to orientation, position, or shape.
	Shape change – Items may switch color and item type (from can to bottle and vice versa). No changes to orientation or position.
	Random item bin – This bin will contain the same items (1 green can, 1 red can and 1 blue bottle, 1 marker and 2 cubes (green and blue)), however the items may change position or orientation.
	Orientation change – Items will remain in the same position but may change orientation. I.e., rotated

Example worlds

The two provided examples show how the layout conditions can affect the unknown environment used for testing the submitted algorithm.

World 1	World 2
	

10 Points	20 Points	30 Points

Getting Started

Simulation Environment

The simulation environment can be found here. The environment runs on MATLAB 2025b and has the following toolbox/support package dependencies

- Simulink 3D Animation
- Robotics System Toolbox
- Computer Vision Toolbox

Detailed instructions are also provided in Appendix A at the end of this document.

Programming the Robot with MATLAB and Simulink

To help you get started, templates on how to control the manipulator using MATLAB and Simulink are provided. While the environment has been updated this year and moved away from gazebo, these may still be of use should you use ROS to communicate with the Simulink environment.

MATLAB and Simulink Learning resources:

- MATLAB Academy (MATLAB/Simulink/Stateflow/Deep Learning Onramps)
- MATLAB and Simulink Robotics Arena
- MathWorks Support for RoboCup (Request complimentary license)
- MATLAB and Simulink Documentation
- Questions on MATLAB and Simulink? roboticsarena@mathworks.com

Scoring Rubric

Submitting Models *(40 points)*	*Points*
All files needed to run algorithm are included in submission. Recommendation: Use a MATLAB or Simulink project, and verify using MATLAB Dependency Analyzer or Simulink Dependency Analyzer	10
Video of algorithm working is submitted.	10
Code is reproducible without any errors.	20
Using Sensors *(80 points)*	
Effective use of RGB camera. Data processed from the RGB camera leads to successful classification of at least one bottle or can.	40
Effective use of depth sensor. Data processed from the depth camera leads to successful classification of at least one bottle or can.	40
Placing Items in Correct bins *(580 points)* *Cans and Spam go in the green bin* *Bottles and Markers go into the blue bin* *Green and Purple cubes go in the Green bin Blue and Red cubes go in the blue bin*	*(Some objects can be different shapes according to environment rules)*
Green Can	10 per can *(3-4 cans available)*
Yellow Can	20 per can *(3-4 cans available)*
Red Can	30 per can *(3 cans available)*
Blue Bottle	10 per bottle *(3-4 bottles available)*
Yellow Bottle	20 per bottle *(3-4 bottles available)*
Red Bottle	30 per bottle *(2 bottles available)*
Cubes	10 per cube (8 cubes available)
Spam	20 per can (*3 available*)
Markers	20 per marker (*3 available*)

Deductions • Item placed in wrong bin: - 10 • Item falls on floor: - 5 • Object moves through a robot part: -5	
Bonus Points (180 points) *Points will be awarded regardless of whether the item is placed in a bin.*	
Robot lifts at least one item from table in the Shape change region (Awarded once only)	10
Robot retrieves at least one item from random item bin (Awarded once only)	20
Single cube is released and then picked up from the weighing scale (Awarded per cube)	15 (per cube, max 120 total)
Minimum Points for Certificate	**60**
Total Possible Points	**880**

Appendix A – Detailed Instructions for Simulation Environment

Getting Started:

- Download and unzip the model from the link received via email
- Open RoboCup_ARM.prj. Note that opening the Simulink model (RoboCup_ARM.slx) directly would not set up the required files and model data correctly.
- Under Simulation Configuration, you can choose between two different simulation worlds. Please uncomment only one of the worlds. By default, World 1 is enabled. To uncomment a block, right-click on the block and choose uncomment.
- Click on Run to run the Simulink model. This should open the simulation environment with the robot picking and dropping a cube. Successful execution of this ensures the model is working as expected. If you run into any issues, please reach out to us.

Model Overview:

The Simulink model by default has three distinct sections.

- Simulation Environment Configuration
- Robot Subsystem
- Planning and Behavior

The Simulation environment contains blocks necessary to setup the simulation world. The Robot Subsystem takes in joint configuration and gripper status, which are then combined before passing it to the robot block. The binary signal target attached returns True when the gripper grabs an object. The other output signals - depth image, RGB image, camera translation, camera rotation and joint configuration can be used in your algorithm.

Note: Please do not change anything in the Simulation Configuration and Robot Subsystem. These build the environment and contain the necessary functions to animate the robot within the environment.

You are free to modify the 'Planning and Behavior' section to solve the sorting problem. The environment outputs RGB image and a depth image from the camera mounted on the robot arm. There is an example trajectory provided which moves the robot to pick and drop a marker.

The robot subsystem takes 2 inputs:

- Robot configuration: A 6x1 joint configuration at each timestep.
- Gripper Status: A binary value that opens or closes the gripper. (0

  = OPEN, 1 = CLOSE)

Submission

Your solution is expected to provide the 2 values (Robot configuration and Gripper status) at each time step. In addition, you can output a stop signal when you have reached the end of the solution. This would stop the simulation which otherwise would run infinitely.

You are allowed to use the full range of MATLAB products to solve this challenge. Should you use another programming language/tool, you are expected to provide all the necessary files for the organizers to run and verify your solution.

For submission, you will be required to provide a public access link to your code and a video of your solution. The submission form will be emailed to the registered participants.

Prohibited Actions

The following list contains some actions which are prohibited and would result in a dismissal of the solution or a heavy point deduction, at the discretion of the judges.

- Accessing object locations or orientation through the environment parameters.
- Altering the lighting or other environmental parameters
- Changing the location/orientation of the camera attached to the manipulator
- Obtaining segmentation data from the camera attached to the manipulator

Any other action that does not comply with the spirit of the competition, which is developing a realistic manipulation algorithm, will be subject to a point reduction or a dismissal at the discretion of the judges.